Stability Analysis of Discrete-Time System

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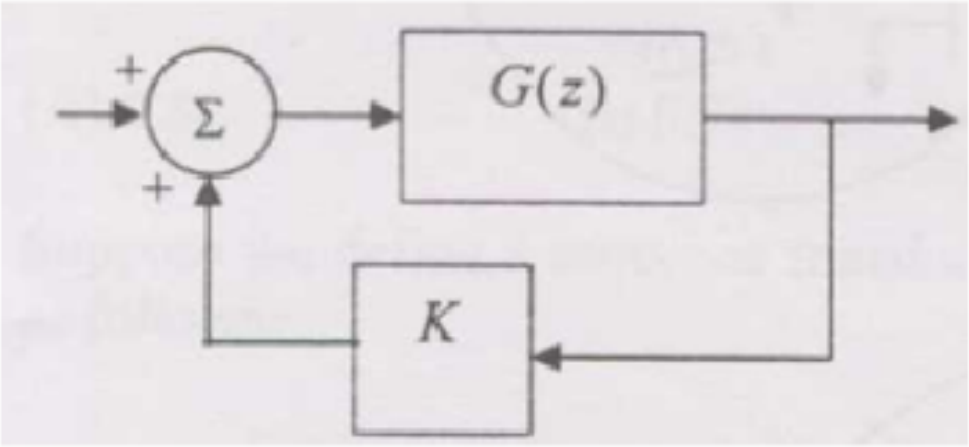
Problem Statement

Consider the discrete-time system shown in the figure where the impulse response $G(z)$ is $g(0) = 0, g(1) = g(2) = 1,\ g(3) = g(4) = ... = 0$



This system is stable for range of values of K

1. $[-1, 1/2]$
2. $[-1, 1]$
3. $[-1/2, 1]$
4. $[-1/2, 2]$

Step 1: Impulse Response to Transfer Function

Impulse response definition:

$$G(z) = \sum g(n) z^{-n}$$

Given $g(1) = 1$ and $g(2) = 1$:

$$G(z) = z^{-1} + z^{-2}$$

Key Idea

z^{-1} represents a one-step delay in the discrete-time domain.

Step 2: Block Diagram Equation

From the feedback loop logic:

$$Y(z) = G(z)\big(X(z) + KY(z)\big)$$

Substitute $G(z)$:

$$Y(z) = (z^{-1} + z^{-2})(X(z) + KY(z))$$

Step 3: Expand the Equation

Expand terms:

$$Y(z) = (z^{-1} + z^{-2})X(z) + K(z^{-1} + z^{-2})Y(z)$$

Rearrange to group $Y(z)$:

$$Y(z) - K(z^{-1} + z^{-2})Y(z) = (z^{-1} + z^{-2})X(z)$$

Factor $Y(z)$:

$$Y(z)\left[1 - K(z^{-1} + z^{-2})\right] = (z^{-1} + z^{-2})X(z)$$

Step 4: Characteristic Equation

For stability, focus on the denominator of the transfer function $H(z) = \frac{Y(z)}{X(z)}$:

$$1 - K(z^{-1} + z^{-2}) = 0$$

Multiply by z^2 to get the standard polynomial form:

$$z^2 - Kz - K = 0$$

Characteristic Polynomial

$$P(z) = z^2 - Kz - K$$

Step 5: Stability Concept

For discrete-time systems:

Stability Condition

All roots of the characteristic equation must satisfy:

$$|z| < 1$$

Instead of solving for roots directly, we apply the **Jury Stability Test**.

Jury Stability Test (2nd Order)

For a second-order polynomial $z^2 + a_1 z + a_0$, the stability conditions are:

1. $|a_0| < 1$
2. $P(1) > 0 \implies 1 + a_1 + a_0 > 0$
3. $P(-1) > 0 \implies 1 - a_1 + a_0 > 0$

All three must be satisfied for the system to be stable.

Step 6: Apply Jury Test

Our polynomial: $z^2 - Kz - K$
Here, $a_1 = -K$ and $a_0 = -K$.
Condition 1: $|-K| < 1 \implies |K| < 1$
Condition 2: $1 + (-K) + (-K) > 0$

$$1 - 2K > 0 \implies K < \frac{1}{2}$$

Condition 3: $1 - (-K) + (-K) > 0$

$$1 > 0 \quad \text{(Always true)}$$

Final Answer

Combining the active constraints:

$$|K| < 1 \quad \text{and} \quad K < \frac{1}{2}$$

This results in the range:

$$\boxed{-1 < K < \frac{1}{2}}$$

Conclusion

The system is stable if and only if K stays within this interval.

Key Takeaways

- z^{-1} functions as a unit delay.
- The characteristic equation determines system behavior.
- Stability requires poles to stay inside the **Unit Circle**.
- The **Jury Test** is a powerful tool for checking stability without calculating complex roots.

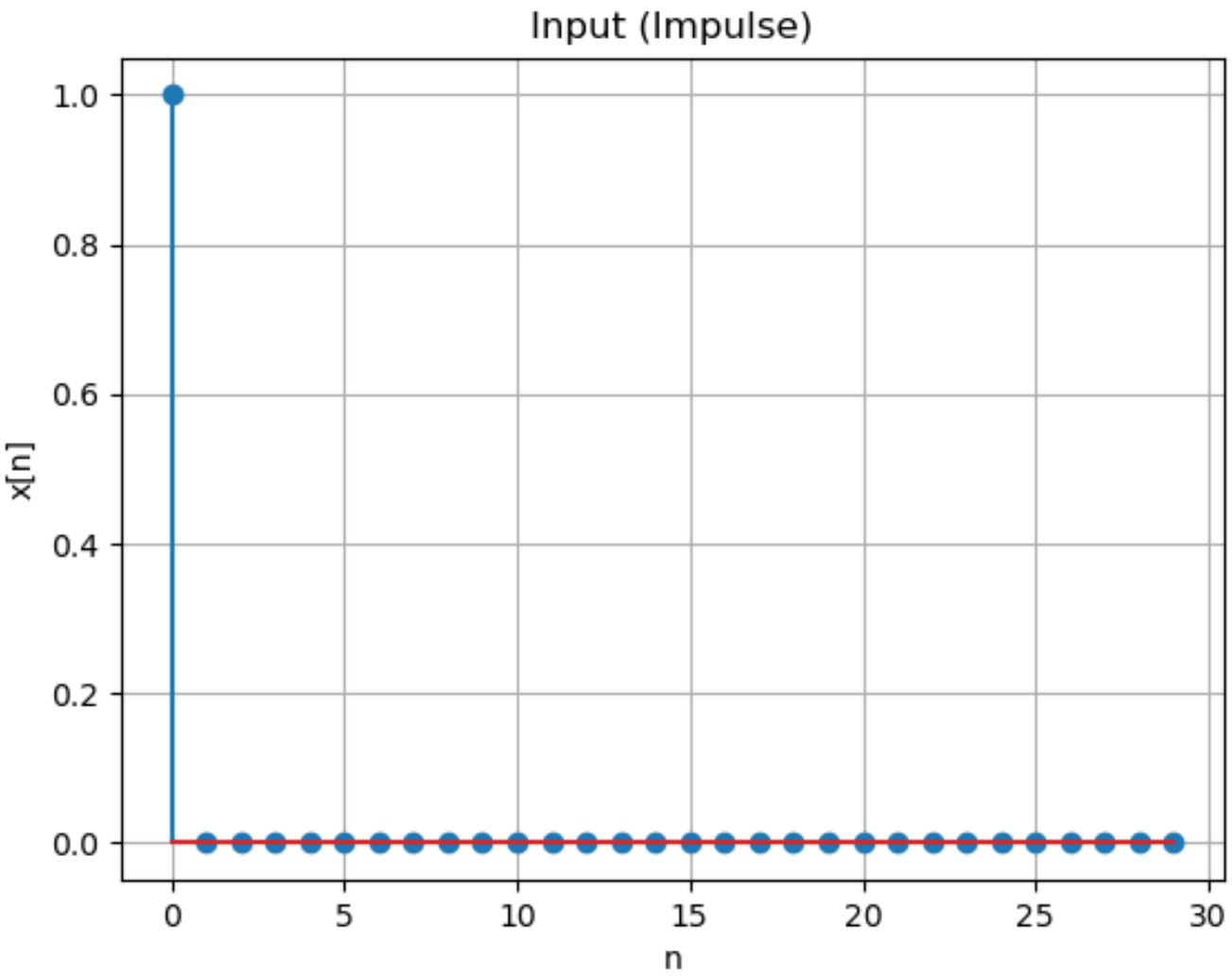
```python
import numpy as np
import matplotlib.pyplot as plt

def simulate_system(K, N=30):
    n = np.arange(N)

    # Input
    x = np.zeros(N)
    x[0] = 1

    # Output
    y = np.zeros(N)

    for i in range(N):
        x1 = x[i-1] if i-1 >= 0 else 0
        x2 = x[i-2] if i-2 >= 0 else 0
        y1 = y[i-1] if i-1 >= 0 else 0
        y2 = y[i-2] if i-2 >= 0 else 0

        y[i] = x1 + x2 + K*y1 + K*y2

```

```
    return n, x, y


def plot_system(K):

    n, x, y = simulate_system(K)

    # impulse response
    g = np.zeros_like(n)
    if len(n) > 1:
        g[1] = 1
    if len(n) > 2:
        g[2] = 1

    # Input plot
    plt.figure()
    plt.stem(n, x)
    plt.title("Input (Impulse)")
    plt.xlabel("n")
    plt.ylabel("x[n]")
    plt.grid()
```

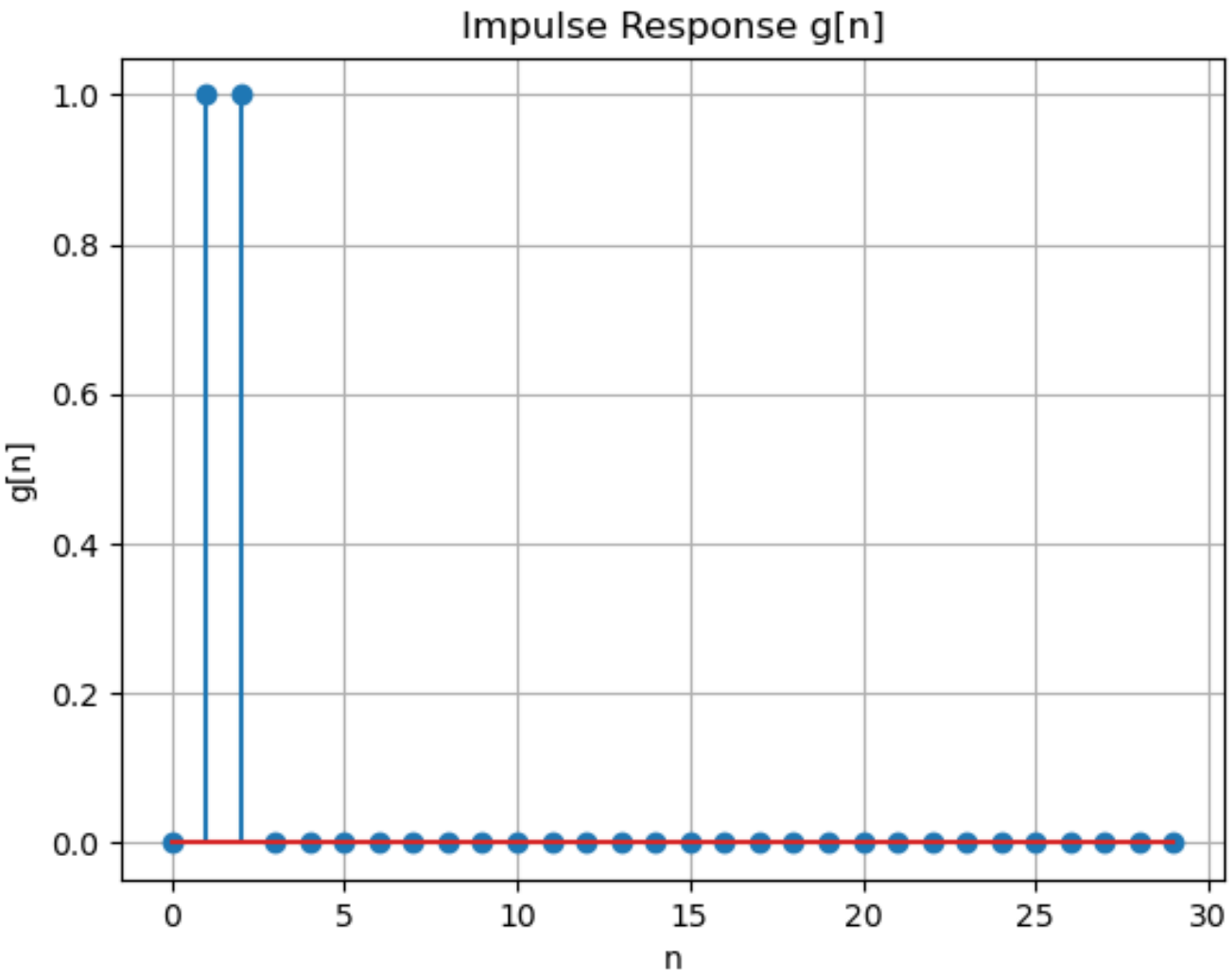
```
plt.savefig("../figs/input.png")
plt.close()

# Impulse response plot
plt.figure()
plt.stem(n, g)
plt.title("Impulse Response g[n]")
plt.xlabel("n")
plt.ylabel("g[n]")
plt.grid()
plt.savefig("../figs/impulse.png")
plt.close()

# Output plot
plt.figure()
plt.stem(n, y)
plt.title(f"Output y[n] (K={K})")
plt.xlabel("n")
plt.ylabel("y[n]")
plt.grid()
plt.savefig("../figs/output1.png")
```

```
    plt.close()


# Run
plot_system(0.7)
```

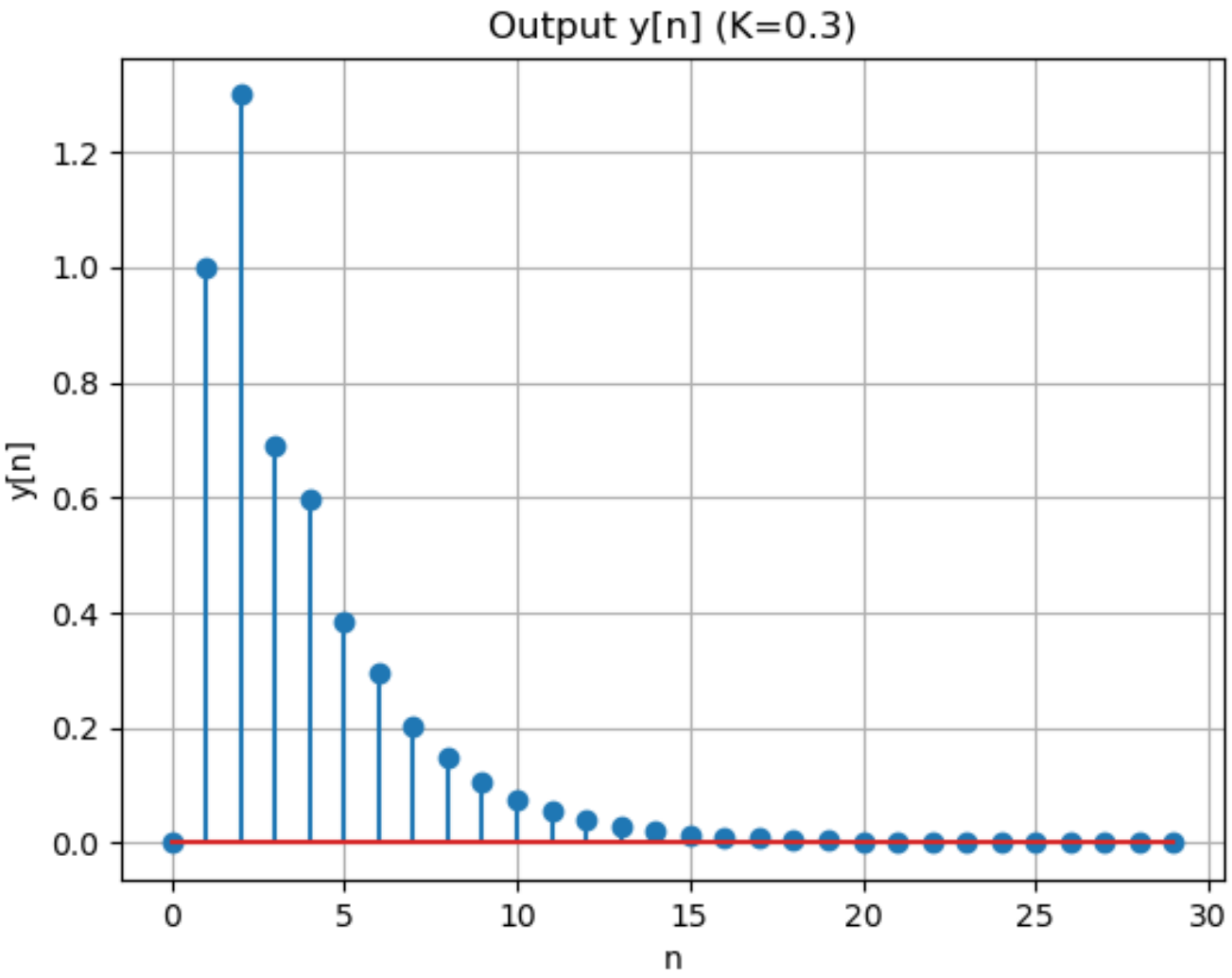
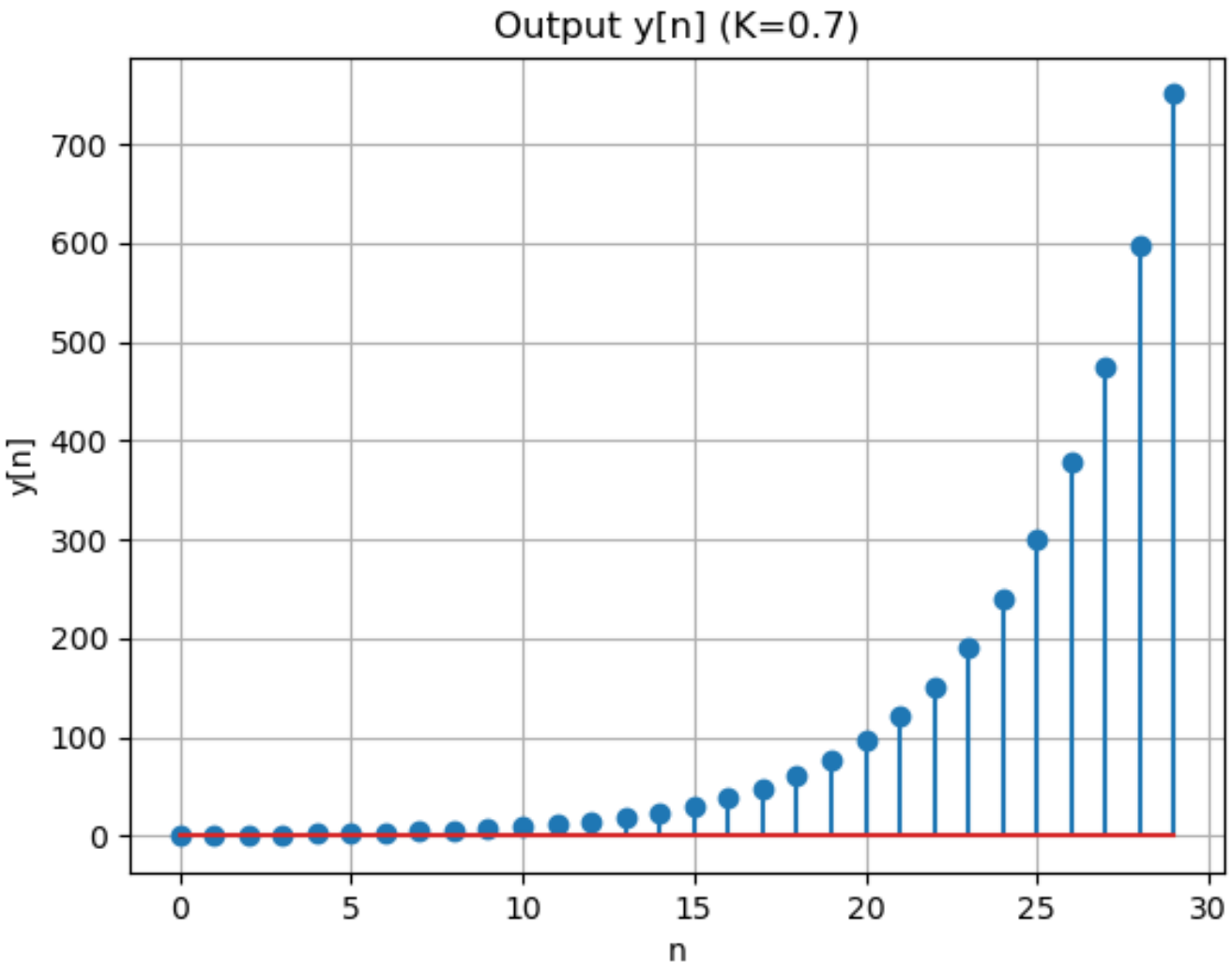
Input Signal



Impulse Response



System Output (Stable Case)



System Output (Unstable Case)